

Standard Shader Ball

A Modern and Feature-Rich Render Test Scene

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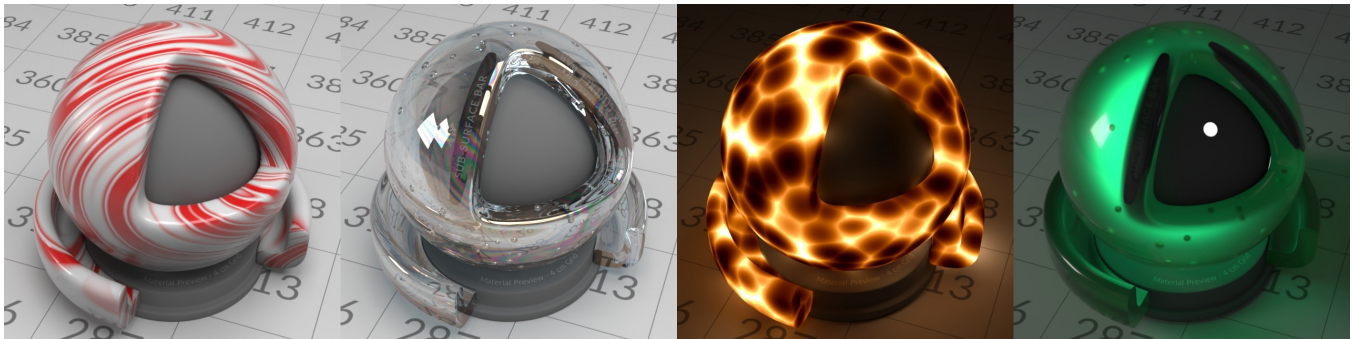


Figure 1: Examples of materials applied to the shader ball scene.

ABSTRACT

We discuss our recent material test scene contribution to the USD Assets Working Group [2022] and enumerate some of its desirable test qualities. The scene employs modern geometry and material standards and has been released to the rendering community under a permissive licence, so that others may benefit from its useful properties, to encourage standardisation, and to remove the need for continuing reinvention.

CCS CONCEPTS

• Computing methodologies → Rendering.

KEYWORDS

Material Response, Light Transport, Rendering, Texturing, Standards, OpenUSD, OpenEXR, MaterialX, ACEScg, Shader Ball

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1 INTRODUCTION

In the field of computer graphics there is a clear need for rendering environments where simulated material responses can be exercised and analysed. Searching for “shader ball” yields a seemingly infinite variety of shapes and designs.

An obvious question to ask is, besides aesthetic considerations, what differentiates them from each other? Are some scenes better than others? Which qualities are desirable? Why are there so many?

Presented here is a from-scratch reimplement of an existing scene that provides a variety of desirable qualities for material evaluation. This new scene has been released to the rendering community under the permissive CC-BY licence [Creative Commons 2013a]. This work employs state-of-the-art scene and material representations and it is hoped to become a de facto standard in the computer graphics community.

2 BACKGROUND

It isn’t known to the authors when the first shader ball render test scene appeared in the computer graphics community, however, countless scenes in this style have been packaged with digital content creation tools [Autodesk Inc. 2019], sold in asset marketplaces [Yot 2018] and authored as personal projects [Eck 2018].

The material test scene provided with Maxwell Render [2004] is exceptional in that it came with a specification that explains a great number of the design choices made [Anagnostou 2021]. However, the proprietary scene format employed and unclear licensing prevented it from finding wide adoption. This changed in 2021 when the scene was rereleased with a CC-BY-SA licence [Creative Commons 2013b] allowing more general use and modification.

Described in the following sections is a scene that borrows heavily, and with permission, from the concepts in the Maxwell scene.

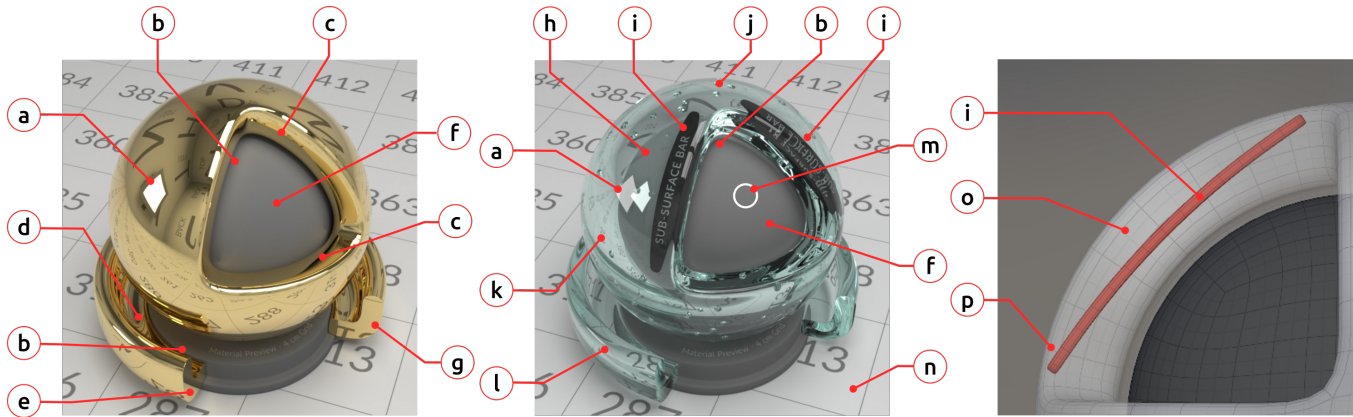


Figure 2: Notable scene features. Left reflective, centre transmissive, and right subsurface configurations. (a): Specular reflectivity and roughness. (b): Caustic receiving areas. (c): Flat areas from triangular cutout. (d): Light trap for recursive reflections. (e): Fresnel checkpoint 63°. (f): Neutral grey core. (g): Fresnel checkpoint 45°. (h): Refracted internal core. (i): Subsurface objects. (j): Air bubbles. (k): Refracted environment. (l): Varying transmission distance. (m): Core emission mode indicator. (n): Grid with unique numbers. (o): Variable thickness above bar. (p): Shallow subsurface region.

3 SCENE PROPERTIES

3.1 Environment Geometry

The test object is enclosed in a box with unique textures for the floor, walls and ceiling, such that reflections and refractions can be traced to particular regions in the scene.

3.2 Material Test Object

The primary surfaces used for material evaluation are a modified sphere and a truncated rim around the base. Secondary diffuse surfaces are also provided to receive various types of indirect illumination from the test material.

Figure 2 shows some of the features exhibited by the test scene. Of particular note are the mixtures of flat and curved surfaces, ray-entrapment zones, and subsurface geometry that in combination exercise a wide variety of material properties.

Furthermore, care has been taken to create a UV-layout that can be used both to exercise spatially varying materials (e.g. surface textures) and also provide a coordinate frame for anisotropic specular roughness.

3.3 External Illumination

Two external emitters are present in the scene. One is to the camera-left side of the render test object with the intention to produce a specular highlight that can be examined for reflectivity and roughness, both isotropic and anisotropic. A second light is placed above the render test object with the intention of creating highlights on and casting caustics from the bevelled edges of the triangular cutout area.

3.4 Internal Illumination

In addition to the transmissive properties exercised by the external lights, the interior geometry of the primary material test objects can be illuminated. Parts of the grey inner core object can be optionally removed to reveal an interior spherical light in three configurations.

3.5 Camera

A camera with known position, orientation and focal length is provided to ensure consistent results across renders.

3.6 Scene Representation

The scene geometry is expressed using OpenUSD [Pixar Animation Studios 2016]. In the original test scene, the object was defined as a highly tessellated mesh that was the result of procedural geometry operations. For our scene we have chosen to use crease-free Catmull-Clark subdivision surfaces, as these provide a much more compact geometric representation and smooth surfaces in the limit. A USD variant containing a triangulated polygonal surface is also provided for rendering engines that do not support subdivision surfaces.

All of our textures are expressed in OpenEXR [1999] files in the industry-standard ACEScsg colour space [2014]. All materials are expressed using MaterialX [Smythe and Stone 2016].

4 LIMITATIONS

USDLux, the mechanism that OpenUSD employs to express light definitions, does not yet support radiometric lights, thus intensity values have been chosen to capture the intent of the original scene.

5 CONCLUSION

We have presented a new test scene to exercise a wide variety of material properties, have used modern scene formats and standards in its implementation and have published it with a permissive licence to the USD Assets Working Group [Mazzone and Rydalch 2023]. It is expected that this will evolve as standards and needs change. We hope that this scene, because of its collection of useful features, will find popularity and bring more consistency to material evaluation across the computer graphics community.

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